

MINUGA KARTHIK KUMAR

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Software engineer specializing in secure systems development, building automated security tooling — vulnerability assessment platforms, deception systems, and web security auditing tools — using Python, FastAPI, and containerized architectures.

Education

Bachelor of Technology – Computer Science & Engineering (AI & ML)
Madanapalle Institute of Technology & Science

2023 – 2027
CGPA: 8.03 / 10.0

Experience

AI / Machine Learning Intern | Viswa Robotic Cars

March 2026 – Present

- Contributing to AI and machine learning projects involving intelligent system development and model implementation.
- Collaborating on Python-based machine learning workflows, gaining real-world software engineering experience.

Google Android Developer Intern | EduSkills × AICTE

March 2026 – June 2026

- Developed Android applications using modern Android application development practices.
- Worked with Android Studio and Git while following structured software development workflows.

Cybersecurity Intern | Cisco Networking Academy

June 2025 – August 2025

- Completed practical training in network security, cyber defense, and cybersecurity fundamentals.
- Gained hands-on exposure to secure networking concepts and threat analysis.

Machine Learning Intern | Viswa Robotic Cars

June 2025 – August 2025

- Worked on machine learning projects involving data preprocessing, model development, and evaluation.
- Collaborated with mentors to understand industry-oriented AI development workflows.

Technical Skills

Programming Languages: Python, JavaScript, C

Backend & Web: Node.js, NestJS, Next.js, REST APIs, JWT Authentication, Flask, Prisma ORM

Databases: PostgreSQL, MongoDB, MySQL, SQLite

Artificial Intelligence: LLM Integration, Tree-sitter, YOLO, Computer Vision, CNNs, DistilBERT, Deep Learning, Model Fine-Tuning

Cybersecurity: Vulnerability Assessment, VAPT, OWASP Top 10, Application Security, Web Security, Attack Surface Analysis, Detection Engineering, Network Security, Threat Hunting, Secure SDLC, DevSecOps, Security Hardening

Security Tools: OWASP ZAP, Nmap, Nikto, Burp Suite, Wireshark, Metasploit, Snort, Suricata, Splunk, ELK Stack, SonarQube, DVWA, pfSense, Autopsy, Kali Linux, Grafana

DevOps & Tooling: Docker, Docker Compose, Nginx, Celery, Redis, Git, VS Code Extension API, Streamlit

Projects

VAPTaaS — Vulnerability Assessment & Penetration Testing Platform

GitHub

Tools | FastAPI | Celery | Redis | PostgreSQL | OWASP ZAP | Nmap | Nikto | React | Claude API

- Architected** an asynchronous vulnerability assessment platform orchestrating reconnaissance (Nmap), web vulnerability scanning (OWASP ZAP), and misconfiguration checks (Nikto) via Celery task queues, containerized with Docker Compose for repeatable security assessments.
- Integrated** the Claude API to auto-generate contextual, developer-readable remediation guidance from raw vulnerability output, translating scan findings into actionable security reporting for engineering teams.
- Implemented** a pytest suite covering scan orchestration and API endpoints, hardening the platform against regressions ahead of deployment.

NexCode — AI-Powered VS Code Extension

GitHub

Tools | TypeScript | Node.js | VS Code Extension API | Tree-sitter | LLM APIs

- Architected** a graph-based code representation engine using Tree-sitter WASM to compress codebase context, cutting LLM token consumption by up to 60% while preserving semantic accuracy.
- Engineered** dependency edge mapping to retain cross-file relationships (imports, calls, class hierarchies) inside a compressed context window, enabling reliable AI reasoning over large codebases.
- Designed** a think → act → observe agent loop with five extensible tools, orchestrating structured, auditable interaction between the LLM and the local development environment.

Distributed Honeytrap Platform

GitHub

Tools | Python | FastAPI | Nginx | PostgreSQL | Grafana | Docker

- Architected** a multi-service deception platform capturing attacker credentials, payloads, source IPs, and behavioral data across independently containerized web, FTP, and API honeypots.
- Built** a normalized event pipeline turning raw activity into structured alerts and attacker profiles, implementing brute-force and scanner detection surfaced through real-time dashboards.

Security Header Analysis Tool

GitHub

Tools | Python | Streamlit | SQLite | ReportLab | Pandas

- Engineered** a security auditing tool evaluating HTTP response headers against OWASP Top 10 categories, flagging missing or misconfigured controls across concurrent, multi-threaded scans of up to 50 domains.
- Built** a scan-history engine with rescan-and-compare analysis and automated, color-coded PDF/JSON reporting for stakeholder-ready remediation.

Crack Detection & Severity Analysis System

GitHub

Tools | Python | YOLO | CNNs | Streamlit

- Engineered** an infrastructure inspection pipeline for automated crack identification and severity classification in civil structures.
- Built** an end-to-end image processing and reporting pipeline, deployed for real-time inspection workflows.

Certifications

- Cyber Security Professional Program (240 Hrs)** — Ethnotech Academy Pvt. Ltd.; coverage of ethical hacking, network defense, web/API security, SOC operations & SIEM, and cloud security fundamentals; hands-on with Burp Suite, OWASP ZAP, Nmap, Metasploit, Splunk, and Wireshark.
- Cybersecurity Essentials & Network Basics — Cisco Networking Academy
- Fundamentals of Cybersecurity — Coursera | Digital 101 Gold Certificate — NASSCOM

Achievements & Leadership

- Winner, Department-Level Project Expo | HackerRank 3-Star Badge in C Programming (Data Structures & Algorithms)
- Event Coordinator, AIMEX 2K25 — managed end-to-end logistics and participant coordination for a departmental technical fest; Coordinator, IQNITE — designed and ran a competitive quiz platform challenge